

Muhammad Usman

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Education

National University of Sciences & Technology, Electrical Engineering Islamabad, Pakistan

- Computer Vision
- Mobile Communications

Experience

Barcelona Supercomputing Center, Research Engineer Barcelona, Spain
2022 – present
4 years

- Enabling BlueField DPU Offloading of MPI Calls in OpenMPI using the underlying UCX framework.
- Ported and evaluated on entire OSU Microbenchmark to prove no performance downgrade.
- Transformed miniWeather application to non-blocking communication and added halo exchange such that a part of compute and node-to-node communication can be offloaded to BlueField DPU to gain 18% speedup using the proposed framework.

ARM (previously DreamBig Semiconductor), Software Engineer Lahore, Pakistan
2020 – 2022
2 years

- Exploring implementation GPUDirect RDMA & GPUDirect Async in RDMA Device Driver Subsystem
- Testing and Documenting steps to reproduce and linux implementation of RoCE with Open vSwitch for Overlay Networks & Link Aggregation using TC Rules

Publications

ODOS-MPI – HPC-Friendly SmartNIC Offloading of Computation/Communication Kernels
Muhammad Usman, Mariano Benito, Sergio Iserte, Antonio J. Peña
dl.acm.org/doi/10.1145/3712285.3759808

DPU offloading programming with the OpenMP API
Muhammad Usman, Mariano Benito, Sergio Iserte, Roger Ferrer, Antonio J. Peña
dl.acm.org/doi/10.1145/3624062.3624165

Skills

Programming

Programming Models

Languages

English
Fluent

Urdu/Hindi
Native Speaker

Spanish
Conversational

Interests

Physics

Certificates

Advanced MPI Programming Nov 2025

CUDA Programming

May 2024

Network Technologies

Oct 2022

References

Antonio J. Peña

Asim Usman